

ROSHAN BALLAV RAUT

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EDUCATION

A. P. Shah Institute of Technology

Kalyan/Thane, India

B.E., Computer Engineering (AI & ML) · Mumbai University

2024 – 2028

Pre-College Scores: SSC 84.6% | HSC 71.33% | MHT-CET 87.9 Percentile

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, SQL

Web / Backend: Flask, REST APIs, HTML, CSS, Jinja2

Databases: PostgreSQL, pgvector (vector search), SQLite

AI / ML: PyTorch (foundations), RAG architecture, Embeddings, Prompt Engineering, LLM APIs (Anthropic, Gemini, Groq)

APIs & Tools: Anthropic API, Google GenAI SDK (Gemini), Alpha Vantage API, Git, GitHub, pytest

Core CS: DSA, Search & Sort Algorithms, Memory Management (Pointers, C), OOP, HTTP Request-Response Cycle, Regular Expressions

PROJECTS

Exactlyl – RAG-Based Document Q&A Web App

CS50x Final Project · Deployed on Render

Flask · PostgreSQL (pgvector) · Gemini Embeddings/Generation · pypdf · PyMuPDF · Gemini Vision

- Built a full-stack RAG Q&A platform: PDF/image ingestion, OCR fallback via PyMuPDF and Gemini Vision, vector embeddings stored in PostgreSQL with pgvector, and Gemini-generated answers grounded in retrieved context.
- Shipped end-to-end and resolved real production issues – deprecated SDK migration, Supabase IPv6/session-pooler constraints, psycopg v3 driver config, and auth refactor from JSON/fetch to form-POST with flash messaging.

Stock Research Agent – Multi-Pass Verification System

Self-directed

Python · Anthropic API (raw) · Alpha Vantage API

- Built an LLM research agent that runs each query three times independently and reconciles the outputs, explicitly flagging numeric discrepancies between passes instead of silently averaging them.
- Designed to reduce single-pass hallucination risk in financial data retrieval and synthesis.

FinHabits – AI-Powered Personal Finance Parser

Built for Google Solution Challenge 2026

Python · React · Groq (Llama 3) · Gemini

- Designed a 3-layer inference pipeline (regex → Groq Llama 3 → Gemini fallback) to parse and categorize Indian bank SMS transactions without manual entry.

MyTaskTracker – Task Tracker with GitHub-Style Heatmap

Deployed on Railway

Node.js · Express · SQLite

- Built and deployed a full-stack task tracker with a GitHub-style contribution heatmap and streak counter to visualize daily activity – live and publicly accessible.

CrisisSync – AI Emergency Coordination Platform (Concept)

Proposed for Google Solution Challenge 2026 · GDG on Campus India

Flutter · Gemini AI · Firebase · Google Maps (planned)

- Designed and prototyped the front-end UI/UX for an AI hotel emergency-response concept: guest SOS flow, command-center dashboard, and floor-level incident map.
- Architected a Gemini-powered triage engine and modified-Dijkstra evacuation-routing approach as the design basis for future backend work; UI-stage only, backend not yet implemented.

COURSEWORK & CURRENTLY EXPLORING

CS50x: Introduction to Computer Science – Harvard University · Final project (Exactlyl) complete

CS50P: Intro to Programming with Python – Harvard University · Final project (Flashcard Quiz CLI) in progress

PyTorch Foundations – hands-on session building and training a circle classifier; deepening into advanced retrieval/embedding systems and production AI serving

CS50AI (Python) – neural networks & deep learning fundamentals